Mathematical and Computational Methods

Lecture 3: Complex numbers II: exponential form, De Moivre, roots

Lecture 2 ended with a striking observation: multiplying complex numbers multiplies their lengths and adds their angles. This lecture turns that observation into a tool. *Euler's formula* packs the polar form into a single exponential $e^{i\theta}$; *De Moivre's theorem* then makes powers immediate; and from powers we obtain the nth roots of any complex number, derive multiple-angle trigonometric identities, and glimpse the kinship between the trigonometric and hyperbolic functions.

Where this fits. We build directly on the polar form and the multiplication rule of Lecture 2, and on the trigonometry of Lecture 1. The one ingredient we take on trust is Euler's formula itself, whose justification through power series belongs to the Calculus module; here we state it and put it to work. The exponential form and De Moivre's theorem then recur throughout the course and across the sciences—oscillations, waves and AC circuits all speak this language.

Contents

Learning objectives

By the end of this lecture you will be able to:

- Move between polar, exponential and trigonometric forms via Euler's formula, and read $e^{i\theta}$ as a rotation.
- State, prove and apply De Moivre's theorem for integer powers.
- Solve $w^n = z$, interpret the roots of unity, and state the fundamental theorem of algebra.
- Derive multiple-angle and trigonometric–hyperbolic identities by complex methods.
- Add two same-frequency oscillations using phasors.

1 Euler's formula and the exponential form

1.1 Euler's formula

The polar form $z = r(\cos\theta + i\sin\theta)$ of Lecture 2 has a remarkably compact rewriting. The complex exponential $e^{i\theta}$ has been given no prior meaning, so we are free to *define* it—and the definition that makes it behave like an exponential is Euler's formula.

Definition 1.1 (The complex exponential $e^{i\theta}$). For every real θ,

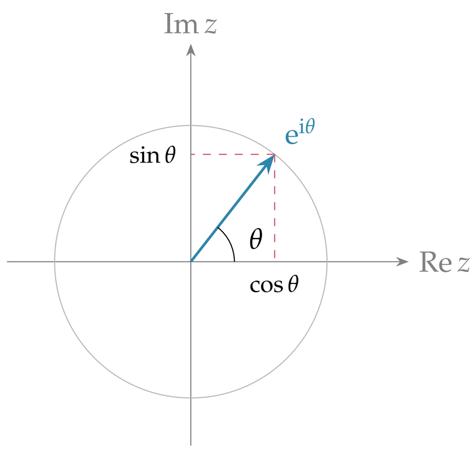
$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Remark. In the Calculus module e^z is instead *defined* by its power series, and Euler's formula becomes a *theorem*, proved by comparing the series of e^x, $\cos\theta$ and $\sin\theta$. Here we have no such machinery, so we take the formula itself as the definition; the checks below confirm it behaves exactly as an exponential should. (For a general exponent, $e^{a+ib} := e^a(\cos b + i\sin b)$.)

Two sanity checks confirm the definition is sensible. First, its modulus is

$$\left|e^{i\theta}\right| = \sqrt{\cos^2\theta + \sin^2\theta} = 1,$$

so $e^{i\theta}$ always lands on the unit circle, at the point making angle θ with the positive real axis.



Second, the law of indices turns into the polar multiplication rule of Lecture 2:

$$\mathrm{e}^{\mathrm{i}\theta}\,\mathrm{e}^{\mathrm{i}\varphi} = (\cos\theta + \mathrm{i}\sin\theta)(\cos\varphi + \mathrm{i}\sin\varphi) = \cos(\theta+\varphi) + \mathrm{i}\sin(\theta+\varphi) = \mathrm{e}^{\mathrm{i}(\theta+\varphi)},$$

using the addition formulas exactly as before. So “adding the arguments” is just “adding the exponents”. The conjugate is $\overline{\mathrm{e}^{\mathrm{i}\theta}} = \mathrm{e}^{-\mathrm{i}\theta} = \cos\theta - \mathrm{i}\sin\theta$.

Example 1.1 ***(Special values and Euler's identity).*** Evaluating at particular angles,

$$\mathrm{e}^{\mathrm{i}\pi/2} = \mathrm{i}, \qquad \mathrm{e}^{\mathrm{i}\pi} = -1, \qquad \mathrm{e}^{2\pi\mathrm{i}} = 1.$$

The middle one rearranges to Euler's celebrated identity $\mathrm{e}^{\mathrm{i}\pi} + 1 = 0$, linking e, i, π, 1 and 0 in a single equation.

1.2 The exponential form of a complex number

Combining $r = |z|$ with Euler's formula gives the third, and most convenient, way to write a complex number.

Definition 1.2 (Exponential form). Every nonzero $z \in \mathbb{C}$ can be written as

$$z = r\,\mathrm{e}^{\mathrm{i}\theta}, \qquad r = |z| > 0, \qquad \theta = \arg(z).$$

This is the same data as the polar form $r(\cos\theta + \mathrm{i}\sin\theta)$, written more compactly.

Conversion is immediate in both directions: read off r and θ exactly as for the polar form (taking the same care over the quadrant, Lecture 2), then write $r\,\mathrm{e}^{\mathrm{i}\theta}$; or expand $r\,\mathrm{e}^{\mathrm{i}\theta} = r\cos\theta + \mathrm{i}\,r\sin\theta$ to return to Cartesian form.

Example 1.2 ***(Moving between forms).*** Since $1+\mathrm{i}$ has modulus $\sqrt{2}$ and argument $\frac{\pi}{4}$,

$$1 + \mathrm{i} = \sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4}.$$

Conversely $2\,\mathrm{e}^{\mathrm{i}\pi/3} = 2\big(\cos\frac{\pi}{3} + \mathrm{i}\sin\frac{\pi}{3}\big) = 2\big(\frac{1}{2} + \mathrm{i}\frac{\sqrt{3}}{2}\big) = 1 + \sqrt{3}\,\mathrm{i}$.

Multiplication and division now look like ordinary index laws—multiply or divide the moduli, add or subtract the exponents:

$$z\,w = rs\,\mathrm{e}^{\mathrm{i}(\theta+\varphi)}, \qquad \frac{z}{w} = \frac{r}{s}\,\mathrm{e}^{\mathrm{i}(\theta-\varphi)}.$$

Example 1.3 ***(Multiplication in exponential form).*** With $z = 1 + \mathrm{i} = \sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4}$,

$$(1+\mathrm{i})^2 = \big(\sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4}\big)^2 = 2\,\mathrm{e}^{\mathrm{i}\pi/2} = 2\mathrm{i},$$

agreeing with the direct expansion $(1+\mathrm{i})^2 = 1 + 2\mathrm{i} + \mathrm{i}^2 = 2\mathrm{i}$.

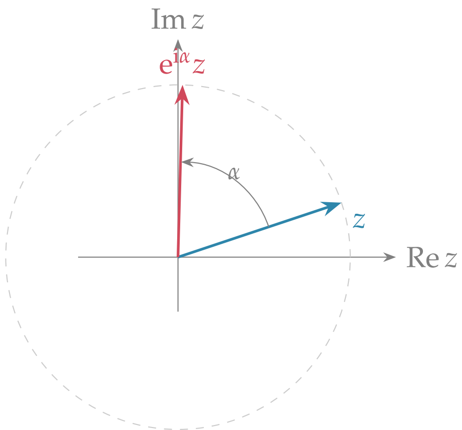
Example 1.4 (*Division in exponential form*). Division is just as quick—divide the moduli and subtract the exponents:

$$\frac{8\,\mathrm{e}^{\mathrm{i}3\pi/4}}{2\,\mathrm{e}^{\mathrm{i}\pi/4}} = \frac{8}{2}\,\mathrm{e}^{\mathrm{i}(3\pi/4-\pi/4)} = 4\,\mathrm{e}^{\mathrm{i}\pi/2} = 4\mathrm{i},$$

and the reciprocal of $z = r\,\mathrm{e}^{\mathrm{i}\theta}$ is simply $1/z = r^{-1}\mathrm{e}^{-\mathrm{i}\theta}$.

1.3 $\mathrm{e}^{\mathrm{i}\theta}$ as a rotation

Because $\left|\mathrm{e}^{\mathrm{i}\alpha}\right| = 1$, multiplying any complex number z by $\mathrm{e}^{\mathrm{i}\alpha}$ leaves its modulus unchanged and increases its argument by α. Geometrically, *multiplication by* $\mathrm{e}^{\mathrm{i}\alpha}$ *rotates* z *anticlockwise through the angle* α *about the origin*—the unit-modulus exponential is the rotation operator of the plane.



Lecture 2's observation that multiplying by i is a quarter turn is the special case $\alpha = \frac{\pi}{2}$, since $\mathrm{i} = \mathrm{e}^{\mathrm{i}\pi/2}$. More generally, multiplying by $w = s\,\mathrm{e}^{\mathrm{i}\alpha}$ both scales by s and rotates by α, which is the geometric content of the multiplication rule.

We can make the rotation completely explicit in real coordinates. Writing $z = x + \mathrm{i}y$ and $\mathrm{e}^{\mathrm{i}\alpha} = \cos\alpha + \mathrm{i}\sin\alpha$, and multiplying out,

$$\mathrm{e}^{\mathrm{i}\alpha}z = (\cos\alpha + \mathrm{i}\sin\alpha)(x + \mathrm{i}y) = (x\cos\alpha - y\sin\alpha) + \mathrm{i}\,(x\sin\alpha + y\cos\alpha).$$

So the new real and imaginary parts (x', y') are obtained from (x, y) by

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

The array of coefficients is the *rotation matrix* of the plane; we meet it again in Unit 6, where its eigenvalues turn out to be exactly $\mathrm{e}^{\pm\mathrm{i}\alpha}$, closing the circle back to the exponential form. As a check, $\alpha = \frac{\pi}{2}$ gives $\cos\alpha = 0$, $\sin\alpha = 1$, so $(x, y) \mapsto (-y, x)$—precisely the multiply-by-i quarter turn of Lecture 2.

1.4 Sine and cosine as exponentials

Euler's formula can be run backwards. Writing it for θ and for $-\theta$,

$$\mathrm{e}^{\mathrm{i}\theta} = \cos\theta + \mathrm{i}\sin\theta, \qquad \mathrm{e}^{-\mathrm{i}\theta} = \cos\theta - \mathrm{i}\sin\theta,$$

and then adding and subtracting gives the *inverse Euler formulas*.

Proposition 1.1 (Cosine and sine as exponentials).

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}, \qquad \sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}.$$

These convert products and powers of sines and cosines into sums of exponentials, which is exactly what is needed both for the multiple-angle identities of Section 3.3 and, later, for integration in the Calculus module.

2 De Moivre's theorem

2.1 Statement and proof

Raising a number in exponential form to a power is effortless, and the result, spelt out in trigonometric form, is De Moivre's theorem.

Theorem 2.1 (De Moivre). For every integer n and every real θ,

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta.$$

Proof. We argue by induction on $n \geq 1$. The base case $n = 1$ is an identity. Assume the result holds for some $n = k \geq 1$. Then

$$(\cos\theta + i\sin\theta)^{k+1} = (\cos\theta + i\sin\theta)^k(\cos\theta + i\sin\theta) = (\cos k\theta + i\sin k\theta)(\cos\theta + i\sin\theta).$$

By the polar multiplication rule of Lecture 2 (both factors have modulus 1, so we simply add the arguments),

$$= \cos(k\theta + \theta) + i\sin(k\theta + \theta) = \cos(k+1)\theta + i\sin(k+1)\theta,$$

which is the statement for $n = k+1$. By induction it holds for all $n \geq 1$. It also holds for $n = 0$ (both sides equal 1), and for negative integers $n = -m$: since $(\cos\theta + i\sin\theta)^{-1} = \cos\theta - i\sin\theta = \cos(-\theta) + i\sin(-\theta)$, raising to the mth power and using the positive case gives $\cos(-m\theta) + i\sin(-m\theta)$. ∎

Remark (The one-line version). In exponential form the theorem is nothing more than the index law

$$\left(e^{i\theta}\right)^n = e^{in\theta},$$

which is precisely why the exponential notation is so powerful: De Moivre's theorem is the statement that $e^{i\theta}$ obeys the usual rules of exponents.

2.2 Integer powers of complex numbers

Combining $z = r\,e^{i\theta}$ with De Moivre gives a clean recipe for any integer power, sidestepping the binomial expansion entirely:

$$z^n = r^n e^{in\theta} = r^n\left(\cos n\theta + i\sin n\theta\right).$$

Example 2.1 ***(A high power without tears).*** Compute $(1+\mathrm{i})^8$. In exponential form $1+\mathrm{i} = \sqrt{2}\,\mathrm{e}^{\mathrm{i}\pi/4}$, so

$$(1+\mathrm{i})^8 = \left(\sqrt{2}\right)^8 \mathrm{e}^{\mathrm{i}\cdot 8\pi/4} = 2^4\,\mathrm{e}^{2\pi\mathrm{i}} = 16.$$

Expanding $(1+\mathrm{i})^8$ by the binomial theorem would take nine terms; De Moivre needs one line.

Example 2.2 ***(A power with a general argument).*** Compute $(\sqrt{3}+\mathrm{i})^6$. Here $\sqrt{3}+\mathrm{i} = 2\,\mathrm{e}^{\mathrm{i}\pi/6}$ (modulus 2, argument $\frac{\pi}{6}$), so

$$(\sqrt{3}+\mathrm{i})^6 = 2^6\,\mathrm{e}^{\mathrm{i}\cdot 6\pi/6} = 64\,\mathrm{e}^{\mathrm{i}\pi} = -64.$$

3 Roots and applications

3.1 The nth roots of a complex number

De Moivre also runs in reverse, to extract roots. An *nth root* of z is any w with $w^n = z$. Writing both in exponential form, $w = \rho\,\mathrm{e}^{\mathrm{i}\phi}$ and $z = r\,\mathrm{e}^{\mathrm{i}\theta}$, the equation $w^n = z$ becomes $\rho^n \mathrm{e}^{\mathrm{i}n\phi} = r\,\mathrm{e}^{\mathrm{i}\theta}$. Matching moduli gives $\rho = r^{1/n}$ (the positive real root), and matching arguments gives $n\phi = \theta + 2\pi k$ for integer k, because the argument is only fixed up to whole turns. Dividing by n and letting k run from 0 to $n-1$ yields n distinct answers.

Theorem 3.1 (The nth roots). For each integer $n \geq 1$, every nonzero $z = r\,\mathrm{e}^{\mathrm{i}\theta}$ has exactly n distinct nth roots,

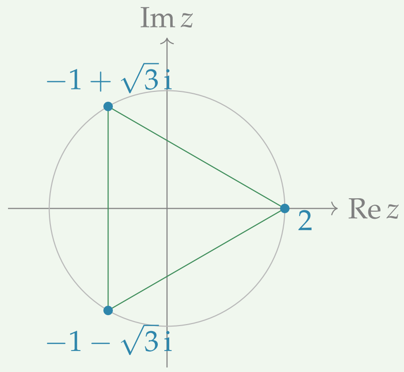
$$w_k = r^{1/n}\,\mathrm{e}^{\mathrm{i}(\theta+2\pi k)/n}, \qquad k = 0, 1, \ldots, n-1.$$

They all have modulus $r^{1/n}$ and their arguments differ by $2\pi/n$, so they sit at the vertices of a regular n-gon on the circle of radius $r^{1/n}$.

Example 3.1 ***(The cube roots of 8).*** Here $z = 8 = 8\,\mathrm{e}^{\mathrm{i}0}$, so $\rho = 8^{1/3} = 2$ and $\phi = \frac{2\pi k}{3}$ for $k = 0, 1, 2$:

$$w_0 = 2, \qquad w_1 = 2\,\mathrm{e}^{\mathrm{i}2\pi/3} = -1 + \sqrt{3}\,\mathrm{i}, \qquad w_2 = 2\,\mathrm{e}^{\mathrm{i}4\pi/3} = -1 - \sqrt{3}\,\mathrm{i}.$$

The familiar real root 2 is joined by a conjugate pair, the three forming an equilateral triangle on the circle of radius 2.



Remark (The roots are rotations of one another). Once a single nth root w_0 is known, every other is w_0 times a power of $e^{i2\pi/n}$:

$$w_k = w_0\, e^{i2\pi k/n}, \qquad k = 0, 1, \ldots, n-1.$$

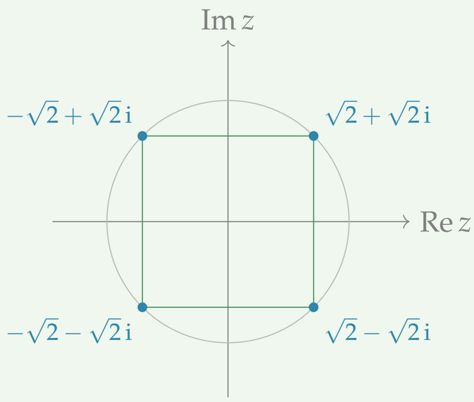
Geometrically the roots are successive rotations of w_0 through $2\pi/n$, which is why they always form a regular polygon.

When the argument of z is not zero, the whole polygon is simply turned to match.

Example 3.2 ***(The fourth roots of*** -16***).*** Write $-16 = 16\,e^{i\pi}$. Then $\rho = 16^{1/4} = 2$ and $\phi = \frac{\pi+2\pi k}{4}$ for $k = 0, 1, 2, 3$, giving arguments $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$ and the four roots

$$\sqrt{2}+\sqrt{2}\,i, \quad -\sqrt{2}+\sqrt{2}\,i, \quad -\sqrt{2}-\sqrt{2}\,i, \quad \sqrt{2}-\sqrt{2}\,i,$$

the corners of a square inscribed in the circle of radius 2, tilted by $\frac{\pi}{4}$.



The two examples above had real right-hand sides; the recipe is no harder when z itself is genuinely complex—one simply reads off its modulus and argument first.

Example 3.3 ***(Cube roots with a slanted argument).*** Solve $w^3 = -2+2i$. First put the right-hand side in exponential form: its modulus is $|-2+2i| = \sqrt{4+4} = 2\sqrt{2}$ and, since it lies in the second quadrant, its argument is $\frac{3\pi}{4}$, so $-2+2i = 2\sqrt{2}\,e^{i3\pi/4}$. The three cube roots therefore have modulus $\rho = (2\sqrt{2})^{1/3} = 2^{1/2} = \sqrt{2}$ and arguments $\phi = \frac{3\pi/4+2\pi k}{3} = \frac{\pi}{4} + \frac{2\pi k}{3}$ for $k = 0, 1, 2$:

$$w_0 = \sqrt{2}\,e^{i\pi/4}, \qquad w_1 = \sqrt{2}\,e^{i11\pi/12}, \qquad w_2 = \sqrt{2}\,e^{i19\pi/12}.$$

The first is the tidy value $w_0 = \sqrt{2}(\cos\frac{\pi}{4} + \mathrm{i}\sin\frac{\pi}{4}) = 1+\mathrm{i}$, and indeed $(1+\mathrm{i})^3 = (1+\mathrm{i})^2(1+\mathrm{i}) = 2\mathrm{i}\,(1+\mathrm{i}) = -2+2\mathrm{i}$, confirming the root. By the rotation remark above the other two are w_0 turned through $\frac{2\pi}{3}$ and $\frac{4\pi}{3}$—that is, w_0 times the cube roots of unity—giving in Cartesian form

$$w_1 = -\tfrac{1+\sqrt{3}}{2} + \tfrac{\sqrt{3}-1}{2}\,\mathrm{i}, \qquad w_2 = \tfrac{\sqrt{3}-1}{2} - \tfrac{1+\sqrt{3}}{2}\,\mathrm{i},$$

the three together forming an equilateral triangle on the circle of radius $\sqrt{2}$.

3.2 Square roots the algebraic way

The exponential method above needs the argument of z in hand. When that argument is a standard angle this is no obstacle, but for a number such as $3+4\mathrm{i}$—whose argument $\arctan\frac{4}{3}$ is not a familiar fraction of π—it is often cleaner to find a square root directly in Cartesian form, by equating real and imaginary parts.

Example 3.4 ***(A square root by equating parts).*** Find the square roots of $3+4\mathrm{i}$. Seek $a+\mathrm{i}b$ with a, b real and $(a+\mathrm{i}b)^2 = 3+4\mathrm{i}$. Expanding the left-hand side,

$$(a+\mathrm{i}b)^2 = a^2 - b^2 + 2ab\,\mathrm{i},$$

so matching real and imaginary parts gives the two real equations

$$a^2 - b^2 = 3, \qquad 2ab = 4.$$

From the second, $b = 2/a$ (note $a \neq 0$). Substituting into the first, $a^2 - 4/a^2 = 3$, and multiplying through by a^2,

$$a^4 - 3a^2 - 4 = 0 \quad\Longrightarrow\quad (a^2-4)(a^2+1) = 0.$$

Since a is real, $a^2+1=0$ is impossible and must be discarded; only $a^2 = 4$ survives, giving $a = \pm 2$ and then $b = 2/a = \pm 1$ with the *same* sign. The two square roots are therefore

$$\sqrt{3+4\mathrm{i}} = \pm(2+\mathrm{i}),$$

and indeed $(2+\mathrm{i})^2 = 4+4\mathrm{i}+\mathrm{i}^2 = 3+4\mathrm{i}$.

Remark (Why two roots, and a shortcut). The two answers are negatives of one another—the $n=2$ case of the general rule that nth roots differ by factors of $\mathrm{e}^{\mathrm{i}2\pi/n} = \mathrm{e}^{\mathrm{i}\pi} = -1$. A useful third equation comes from moduli: taking $|\cdot|^2$ of $(a+\mathrm{i}b)^2 = 3+4\mathrm{i}$ gives $a^2+b^2 = |3+4\mathrm{i}| = 5$. Adding and subtracting this with $a^2 - b^2 = 3$ solves the squares directly—$a^2 = 4$, $b^2 = 1$—and the sign of $2ab = 4 > 0$ then fixes a, b with equal signs. This modulus trick is the quickest route for a general square root.

3.3 The roots of unity

The most important special case takes $z = 1$: the solutions of $w^n = 1$ are the *nth roots of unity*.

Definition 3.1 (Roots of unity). The nth roots of unity are

$$w_k = \mathrm{e}^{\mathrm{i}2\pi k/n}, \qquad k = 0, 1, \ldots, n-1.$$

Writing $\omega = \mathrm{e}^{\mathrm{i}2\pi/n}$, they are the powers $1, \omega, \omega^2, \ldots, \omega^{n-1}$: n equally spaced points on the unit circle, forming a regular n-gon with one vertex at 1.

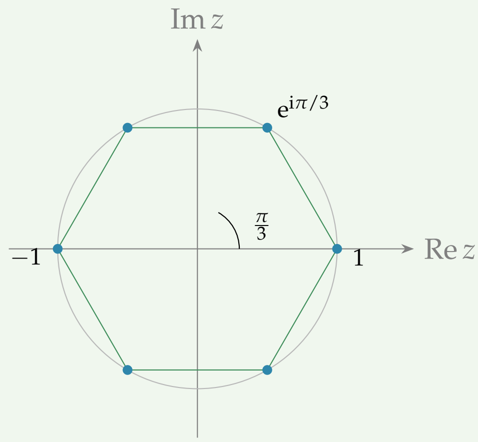
Proposition 3.1 (Their sum vanishes). For $n \geq 2$ the nth roots of unity sum to zero, $1 + \omega + \omega^2 + \cdots + \omega^{n-1} = 0$.

Proof. This is a geometric series with ratio $\omega \neq 1$, so it equals $\dfrac{\omega^n - 1}{\omega - 1} = \dfrac{1-1}{\omega - 1} = 0$, since $\omega^n = 1$. ■

Example 3.5 ***(The sixth roots of unity).*** With $n = 6$ the roots are $\mathrm{e}^{\mathrm{i}\pi k/3}$ for $k = 0, \ldots, 5$:

$$1, \quad \tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\mathrm{i}, \quad -\tfrac{1}{2} + \tfrac{\sqrt{3}}{2}\mathrm{i}, \quad -1, \quad -\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\mathrm{i}, \quad \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\mathrm{i},$$

the vertices of a regular hexagon. Notice they occur in conjugate pairs and include both real roots ± 1.



Remark (Roots of unity in data science and pure mathematics). The evenly spaced roots of unity are not just decorative. Sampling a signal at the nth roots of unity is the basis of the *discrete Fourier transform*—the powers ω^k are exactly the frequencies a length-n DFT measures, and the fast algorithm (the FFT) that data scientists and engineers use to analyse audio, images and time series rests on their arithmetic. For the pure mathematician, the same set $\{1, \omega, \omega^2, \ldots, \omega^{n-1}\}$ is the simplest example of a *cyclic group*: every root is a power of the single generator ω.

3.4 The fundamental theorem of algebra

The equation $w^n = z$ solved in §3.1 is, after rearranging, the polynomial equation $w^n - z = 0$ of degree n, and we found it has exactly n solutions in $\mathbb{C}$—no more and no fewer. This is the simplest case of the central fact about polynomials over the complex numbers.

Theorem 3.2 (Fundamental theorem of algebra). A polynomial of degree $n \geq 1$ with complex coefficients,

$$p(w) = a_n w^n + a_{n-1} w^{n-1} + \cdots + a_1 w + a_0 \qquad (a_n \neq 0),$$

has exactly n roots in $\mathbb{C}$, counted with multiplicity. Equivalently, it factors completely into linear factors,

$$p(w) = a_n(w - w_1)(w - w_2) \cdots (w - w_n).$$

Remark. A genuine proof needs tools from the analysis side of mathematics and is beyond our scope. The theorem is the deepest reason the chain of number systems $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C}$ stops at $\mathbb{C}$: unlike $\mathbb{R}$, the complex numbers are *algebraically closed*—no polynomial equation can lead outside them, so no further extension is ever needed. The nth roots of z above are one illustration; the conjugate-pair roots of the real quadratic in Lecture 2 are another, reflecting the general rule that a polynomial with *real* coefficients has its non-real roots in conjugate pairs. That rule is worth proving, since it costs only the conjugation algebra of Lecture 2.

Proposition 3.2 (Real polynomials have conjugate-pair roots). Let $p(w) = a_n w^n + \cdots + a_1 w + a_0$ have *real* coefficients $a_0, \ldots, a_n$. If w is a root, so is $\overline{w}$.

Proof. Conjugation respects sums and products and fixes every real number, so each $\overline{a_k} = a_k$ and $\overline{w^k} = \overline{w}^k$. Hence

$$p(\overline{w}) = \sum_{k=0}^{n} a_k \, \overline{w}^k = \sum_{k=0}^{n} \overline{a_k} \, \overline{w^k} = \overline{\sum_{k=0}^{n} a_k w^k} = \overline{p(w)}.$$

If $p(w) = 0$ then $p(\overline{w}) = \overline{0} = 0$, so $\overline{w}$ is a root too. ∎

Thus the non-real roots of a real polynomial come in conjugate pairs $w, \overline{w}$; in particular a real polynomial of *odd* degree, having an odd number of roots, must have at least one real one. This is exactly the conjugate pair seen for the real quadratic of Lecture 2, now established for any degree.

3.5 Multiple-angle identities

De Moivre is also the quickest route to identities for $\cos n\theta$ and $\sin n\theta$. The idea is to expand $(\cos\theta + \mathrm{i}\sin\theta)^n$ with the binomial theorem from school, $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$, then equate it to $\cos n\theta + \mathrm{i}\sin n\theta$ and compare real and imaginary parts.

Example 3.6 *($\cos 3\theta$ **and** $\sin 3\theta$)***.*** Expanding the cube,

$$(\cos\theta + \mathrm{i}\sin\theta)^3 = \cos^3\theta + 3\mathrm{i}\cos^2\theta\sin\theta - 3\cos\theta\sin^2\theta - \mathrm{i}\sin^3\theta.$$

By De Moivre this equals $\cos 3\theta + \mathrm{i}\sin 3\theta$. Comparing real and imaginary parts, then using $\sin^2\theta = 1 - \cos^2\theta$ and $\cos^2\theta = 1 - \sin^2\theta$,

$$\cos 3\theta = \cos^3\theta - 3\cos\theta\sin^2\theta = 4\cos^3\theta - 3\cos\theta,$$

$$\sin 3\theta = 3\cos^2\theta\sin\theta - \sin^3\theta = 3\sin\theta - 4\sin^3\theta.$$

The inverse Euler formulas give the reverse direction—powers of cos or sin written as sums of multiple angles—which is invaluable for integration later.

Example 3.7 ***(A power as a sum of angles).*** Using $\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$,

$$\cos^2\theta = \frac{(e^{i\theta} + e^{-i\theta})^2}{4} = \frac{e^{2i\theta} + 2 + e^{-2i\theta}}{4} = \frac{2\cos 2\theta + 2}{4} = \frac{1 + \cos 2\theta}{2}.$$

A similar calculation gives $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$. Cubing instead, with $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$,

$$\sin^3\theta = \left(\frac{1}{2i}\right)^3 \left(e^{i\theta} - e^{-i\theta}\right)^3, \qquad \left(e^{i\theta} - e^{-i\theta}\right)^3 = e^{3i\theta} - 3e^{i\theta} + 3e^{-i\theta} - e^{-3i\theta}.$$

Pairing the outer and inner exponentials into sines— $e^{3i\theta} - e^{-3i\theta} = 2i\sin 3\theta$ and $e^{i\theta} - e^{-i\theta} = 2i\sin\theta$—the bracket becomes $2i\sin 3\theta - 3\,(2i\sin\theta)$. Dividing by $(2i)^3 = -8i$ (so the factors of i cancel, since $i^3 = -i$),

$$\sin^3\theta = \frac{2i\sin 3\theta - 6i\sin\theta}{-8i} = \tfrac{1}{4}\left(3\sin\theta - \sin 3\theta\right).$$

The odd power gives a combination of *sines*, the natural counterpart of the $\cos^2\theta$ reduction above.

3.6 A link to the hyperbolic functions

The inverse Euler formulas also explain the family resemblance, noted in Lecture 1, between the trigonometric and hyperbolic functions: each pair is the other viewed along the imaginary direction in $\mathbb{C}$. Feeding an imaginary argument into cos and sin,

$$\cos(ix) = \frac{e^{i(ix)} + e^{-i(ix)}}{2} = \frac{e^{-x} + e^{x}}{2} = \cosh x,$$

$$\sin(ix) = \frac{e^{i(ix)} - e^{-i(ix)}}{2i} = \frac{e^{-x} - e^{x}}{2i} = -\frac{e^{x} - e^{-x}}{2i} = i\sinh x,$$

the last step using $-1/i = i$. Running the same calculation the other way—putting an imaginary argument into cosh and sinh—gives the converse pair

$$\cosh(ix) = \frac{e^{ix} + e^{-ix}}{2} = \cos x, \qquad \sinh(ix) = \frac{e^{ix} - e^{-ix}}{2} = i\sin x.$$

So cosh is “cosine along the imaginary axis” and sinh is “sine along the imaginary axis”. This is also why the hyperbolic identities mirror the trigonometric ones: for instance $\cosh^2 x - \sinh^2 x = 1$ is exactly $\cos^2\theta + \sin^2\theta = 1$ read at $\theta = ix$.

3.7 Application: phasors and AC analysis

Remark (For the physics students). In oscillations and alternating-current circuits a sinusoid $A\cos(\omega t + \varphi)$ is written as the real part of a rotating complex exponential,

$$A\cos(\omega t + \varphi) = \mathrm{Re}\big(A\,e^{i\varphi}\,e^{i\omega t}\big) = \mathrm{Re}\big(P\,e^{i\omega t}\big),$$

where the constant complex number $P = A\,e^{i\varphi}$—the *phasor*—stores the amplitude and phase. Because differentiating in t merely multiplies $e^{i\omega t}$ by $i\omega$, calculus with sinusoids becomes algebra with complex numbers.

The single most useful consequence is the painless addition of oscillations of the same frequency—sums that appear throughout AC circuit theory, wave interference and vibrations, and that are tedious to handle with trigonometric identities alone. Since Re is linear and every term carries the same factor $e^{i\omega t}$,

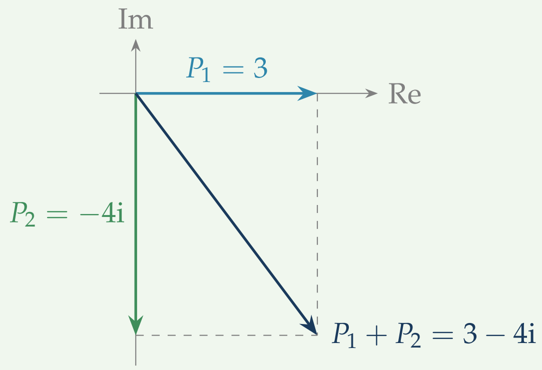
$$\sum_k A_k \cos(\omega t + \varphi_k) = \mathrm{Re}\Big(\big(\textstyle\sum_k P_k\big) e^{i\omega t}\Big),$$

so the result is *another* sinusoid of the same frequency, whose amplitude and phase are the modulus and argument of the phasor sum $\sum_k P_k$. That sum is just vector addition of the phasors in the Argand plane (Lecture 2).

Example 3.8 ***(Combining two oscillations).*** Add $3\cos\omega t + 4\sin\omega t$. Writing $4\sin\omega t = 4\cos(\omega t - \frac{\pi}{2})$, the two phasors are $P_1 = 3\,e^{i0} = 3$ and $P_2 = 4\,e^{-i\pi/2} = -4i$, so

$$P_1 + P_2 = 3 - 4i = 5\,e^{-i\alpha}, \qquad \alpha = \arctan\tfrac{4}{3} \approx 0.927.$$

Therefore $3\cos\omega t + 4\sin\omega t = 5\cos(\omega t - \alpha)$: a single oscillation of amplitude 5, lagging by α. The amplitude is nothing but the hypotenuse of the 3–4–5 right triangle formed by the phasors.



Summary

Concept	Key idea
Euler's formula	$e^{i\theta} = \cos\theta + i\sin\theta$; a unit-circle point; $e^{i\theta}e^{i\varphi} = e^{i(\theta+\varphi)}$ recovers the polar multiplication rule.
Exponential form	$z = r\,e^{i\theta}$; multiply/divide by adding/subtracting exponents; $e^{i\pi} + 1 = 0$.
Rotation	Multiplying by $e^{i\alpha}$ rotates by α (modulus fixed); by $s\,e^{i\alpha}$, scale by s and rotate by α.
Inverse Euler	$\cos\theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$, $\sin\theta = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$.
De Moivre	$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta$ for $n \in \mathbb{Z}$; proved by induction; equivalently $(e^{i\theta})^n = e^{in\theta}$.
Integer powers	$z^n = r^n(\cos n\theta + i\sin n\theta)$.
nth roots	$w_k = r^{1/n}e^{i(\theta+2\pi k)/n}$, $k = 0, \ldots, n-1$: a regular n-gon on the circle of radius $r^{1/n}$.
Roots of unity	$e^{i2\pi k/n}$; powers of $\omega = e^{i2\pi/n}$; they sum to 0.
Fund. thm. algebra	Every degree-n complex polynomial has exactly n roots in $\mathbb{C}$ (with multiplicity); $\mathbb{C}$ is algebraically closed.
Multiple angles	Expand $(\cos\theta + i\sin\theta)^n$ and compare parts (e.g. $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$); reverse with inverse Euler.
Hyperbolic link	$\cos(ix) = \cosh x$, $\sin(ix) = i\sinh x$, $\cosh(ix) = \cos x$, $\sinh(ix) = i\sin x$.
Phasors	Represent $A\cos(\omega t + \varphi)$ by $P = A\,e^{i\varphi}$; add same-frequency oscillations by adding phasors.

Next lecture: a change of subject—vectors, the first step into the linear-algebra strand, with their algebra, the dot and cross products, and lines and planes.

Companion material: a separate *Exercise Sheet 3* (with solutions) develops the practice problems for this unit.